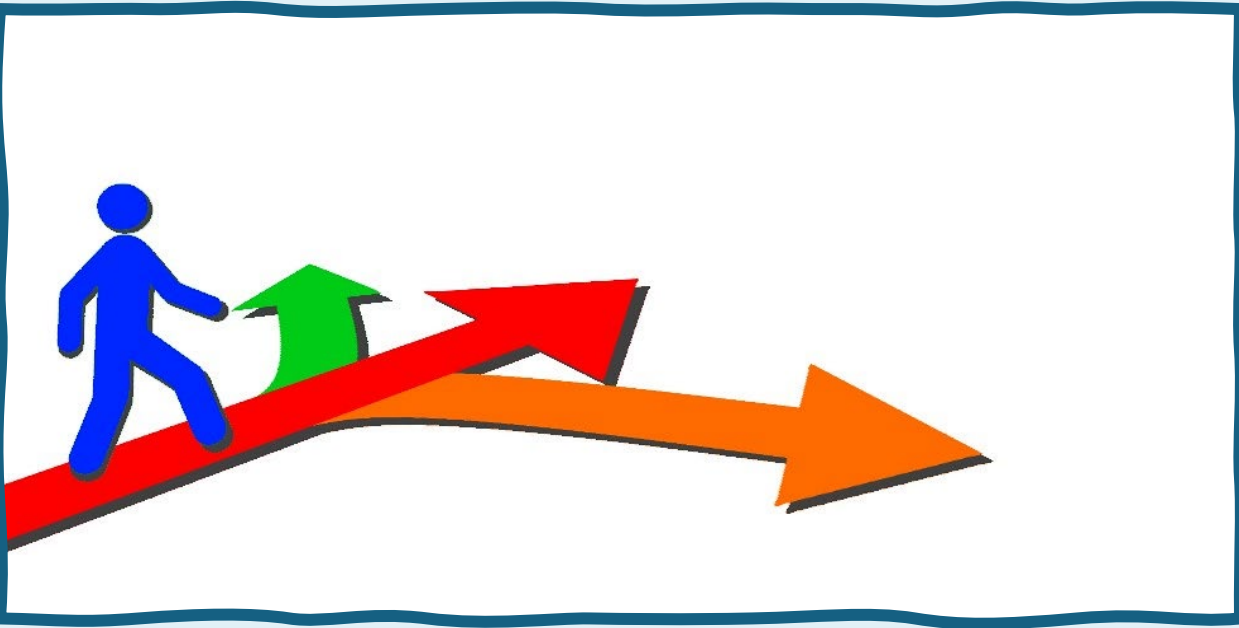


AI Agent Analysis and Probabilistic Modeling- Theory Report Project

Presented by team Skynet

Agent Type

Rational Or Irrational?



Rational, due to the agent choosing the best action based on the history of events and circumstances.

Agent Type (MAI Classification)

- Reflex Agent
- Model Based Agent
- Goal Based Agent
- Utility Based Agent
- Learning Agent



- Not Reflex Agent
 - Due to it needing to keep memory of events and the complexity of the system.
- Not Learning Agent
 - Due to the system being unable to change the algorithm or distinction tree while it is running.



- Model Based Agent
 - Due to it using a model of the world and everything in it.
- Goal Based Agent
 - Due to it having the goal of providing the driver of the car with the most money
- Utility Based Agent
 - Due to it having access to the resources the car has such as the current amount of money.

Why both goal and utility?

The agent needs to be able to find the best action in its current state as well as plan and take multiple steps toward the plan.

Criteria for Performance

- Amount of utility that a car can achieve
- The amount of budget/time per utility
- The performance when there are a lot of vehicles on the road
- Reducing complexity for getting optimal solution
- Reducing uncertainty and inconsistency with results
- Road coverage and crowdsources coverage
- How well it does on incomplete knowledge

Based on these criteria, the agent does far better than greedy and random approaches.



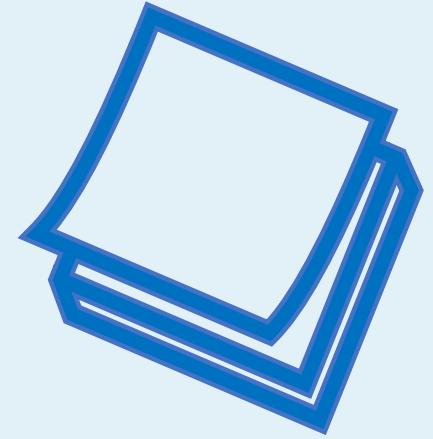
Task Environment Classification

- Fully Observable
- Stochastic
- Sequential
- Dynamic
- Discrete
- Multi-agent
- Competitive



Knowledge Representation

What does our AV need to know?



- The starting/ending location
- The shortest path between them
- The amount of data the AV can collect through its sensors
- The information from the valid adjacent cells and crowdsourcers
- The memory buffer to avoid returning to already-visited places.



Expected Utility Equation and Decision Making

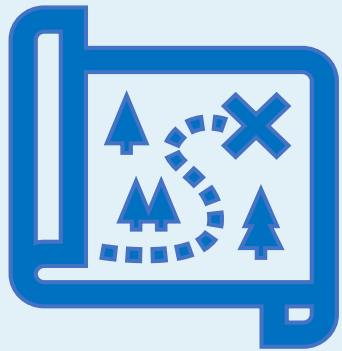


Using the expected utility, the AV can decide which of the available neighboring regions it can travel to that contains the highest value, maximizing the profit on its path.

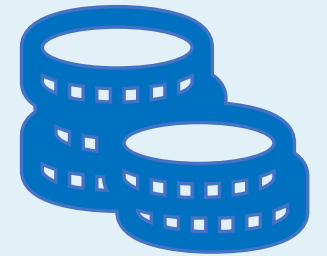
$$EU_i = \sum_{A \subset V, i \in A} \prod_{j \in A} P_j \times \prod_{j \notin A} (1 - P_j) \times u_i | A$$

Expected Sensing Plan and Movement Decisions

The Expected Sensing Plan can impact the AV's route by determining whether cells with the highest expected utility are worth traveling to based on the AV's sensing capacity being greater than or equal to the calculated ESP for that cell.



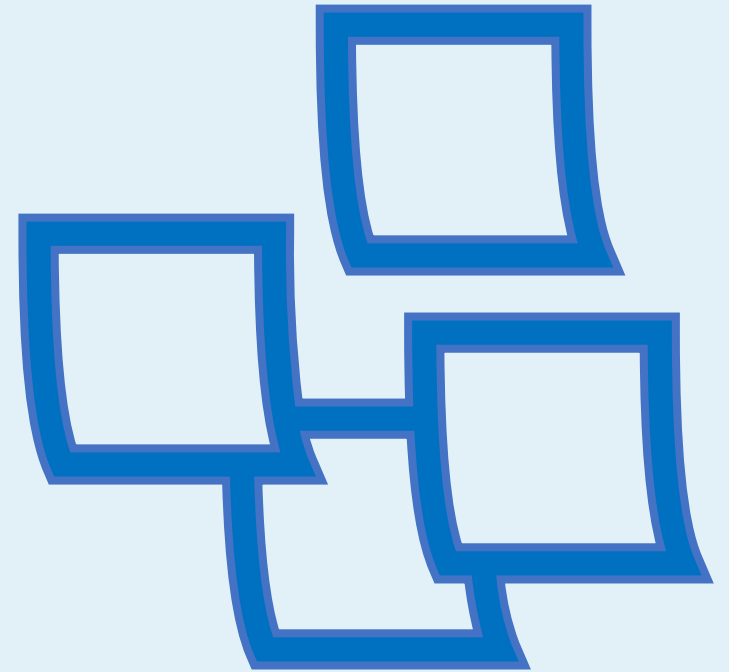
$$ESP_i = \sum_{A \subset V, i \in A} \prod_{j \in A} P_j \times \prod_{j \notin A} (1 - P_j) \times t_i | A.$$



If the AV cannot accommodate the ESP of a neighboring cell in its capacity, it chooses another location where it can collect data.

Characteristics of the set of vehicles' trajectories

- The Approximated Trajectory Nash Equilibrium
 - The AVs maintain a stable configuration and cannot gain any profit by changing their course.
- Deviating from the shortest path to visit areas not on the original path to collect data
- Heading straight for their final destination after reaching their sensing capacity.



Numerical Reasoning

Numerical Reasoning

Expected Utility: An AV (me AV1) considers moving to a cell with reward $R = 120$. Two other AVs may also go there with probabilities of 0.6 and 0.3, respectively. Compute the expected utility of AV1 to visit the cells that contain R . Assume that if AV1 arrives there alone, its utility will be 120 (the whole reward)

To find the expected utility we need to find all possible scenarios where AV1 visits the cell.
Likelihoods:

AV1 is alone: $(1-0.6)*(1-0.3)= 0.28$

AV1 with AV2 only: $0.6*(1-0.3)= 0.42$

AV1 with AV3 only: $(1-0.6)*0.3=0.12$

All three AVs: $0.6*0.3=0.18$

The reward R is decided by the number of agents present in each scenario

AV1 alone: $120/1=120$

AV1 with AV2: $120/2=60$

AV1 with AV3: $120/2=60$

All three AVs: $120/3=40$

The expected utility is the sum of the utility of each outcome multiplied by its probability

$EU = (0.28 * 120) + (0.42 * 60) + (0.12 * 60) + (0.18 * 40) = 73.2$

The expected utility for AV1 is 73.2

Numerical Reasoning

Capacity Constraint: An AV has sensing capacity $b = 8$. Expected sensing plans for upcoming cells are [3, 4, 5]. Which cells can it visit before exhausting capacity?

sensing capacity = 8

Cell 1: 3 units ($8 - 3 = 5$)

Cell 2: 4 units ($5 - 4 = 1$)

Cell 3: 5 units (the AV cannot complete the third cell because the 5 units $>$ 1 remaining sensing capacity.)

Therefor the AV cannot complete visit the first two cells before exhausting capacity.

Numerical Reasoning

Sensing Game Intuition: If more AVs arrive at a crowdsourcer, how does that affect (a) individual sensing plan and (b) individual utility? Explain conceptually.

a) as more AVs arrive at the same crowdsourcer each AV usually contributes less sensing effort because the total required sensings is spread across more participants

b) individual utility generally decreases as more AVs arrive because the reward is divided among more agents.

Conclusion: While the work per AV may decrease so will the reward per AV.

Probability Modeling

Example of Binomial Distribution

- $n = 5$ nearby vehicles
- each independently moves to a cell with probability $p = 0.4$
- what is the probability exactly 2 of them go there?

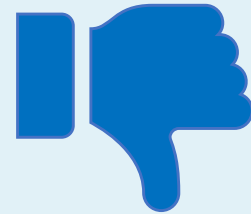
$$P(X = 2) = \binom{5}{2} (0.4)^2 (0.6)^{(5-2)}$$

$$\binom{5}{2} = (5 * 4) / (2 * 1) = 10$$

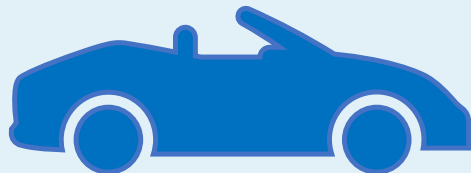
$$P(X = 2) = 10 * 0.16 * 0.216 = 0.3456$$

The Binomial Distribution in the Report

The situation in the report does not follow binomial distribution because there isn't a simple yes or no.

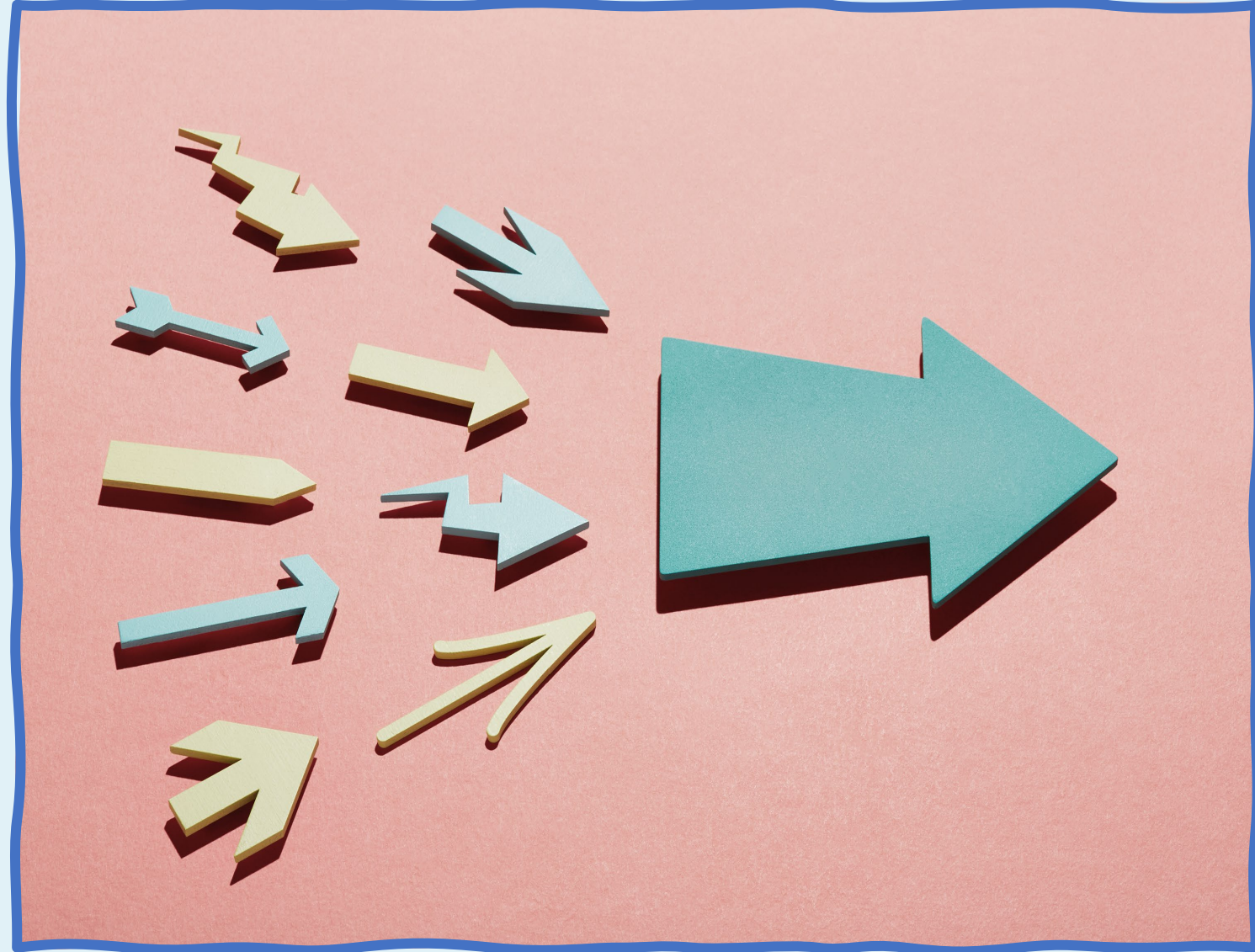


It follows Poisson-Binomial distribution, since there are several different agents who all have different probabilities of participating.



Why Poisson-Binomial?

- Each Instance has different probabilities
- Reduces computational complexity by generalizing the calculations of all instances
- It allows for rapid calculation of probabilities no matter how many agents there are



Combinatorial Complexity

The problem

The problem uses multiple crowdsourcers where their $C = \{c1, c2, cF\}$ and participants $V = \{v1, v2, vM\}$, each with a starting point, $S = \{s1, s1, sM\}$, and a destination, $D = \{d1, d2, dM\}$.

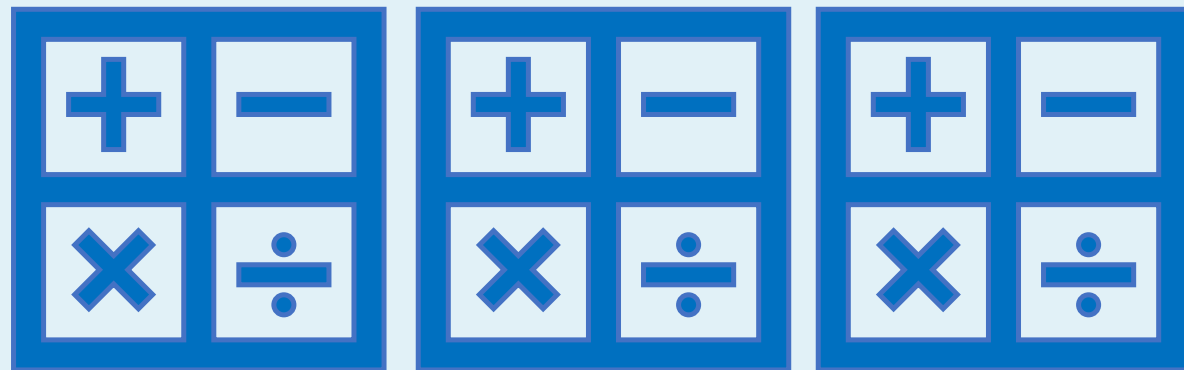
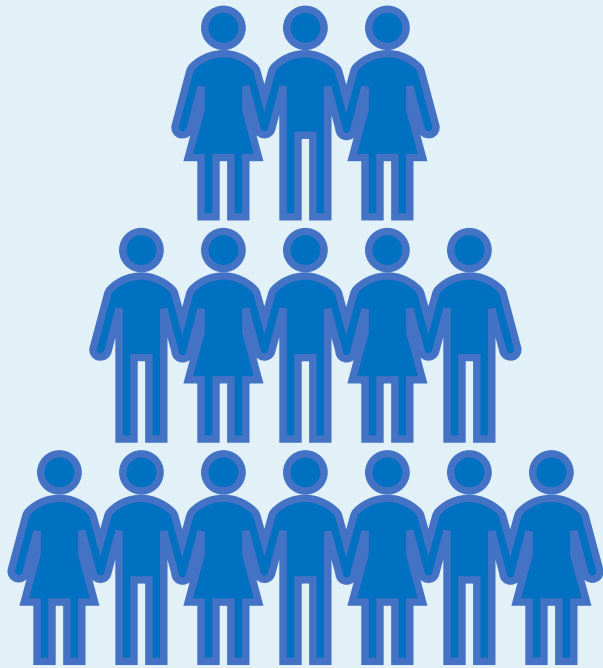
Each crowdsourcer has a reward and destination sub-region. As the number of crowdsourcers goes up, the number of strategy combinations also increases.

$$w_i = \frac{r_i}{\sum_{i \in W_j} r_i} R_j - k_i r_i$$

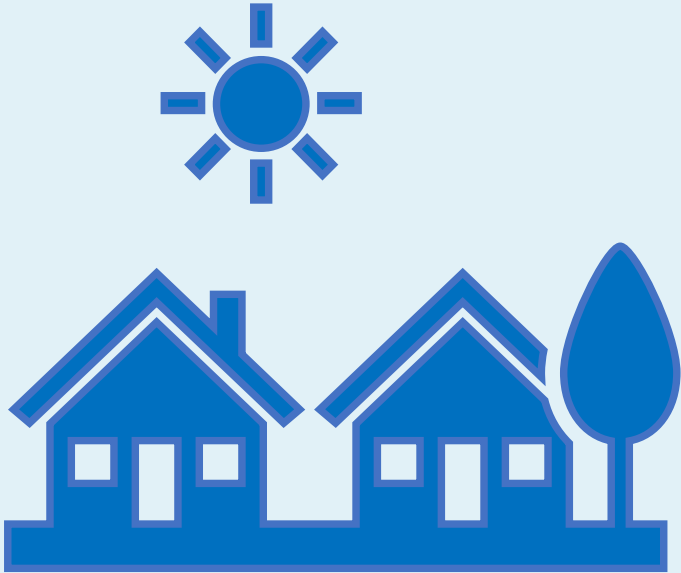
Why not just use the Nash equilibrium?

Finding the exact Nash equilibrium is intractable because of the computational complexity.

As the number of agents goes up or the number of trajectories that an agent can take go up, the number of different strategy combinations follows suit and with that it would take too long to find one for every situation.

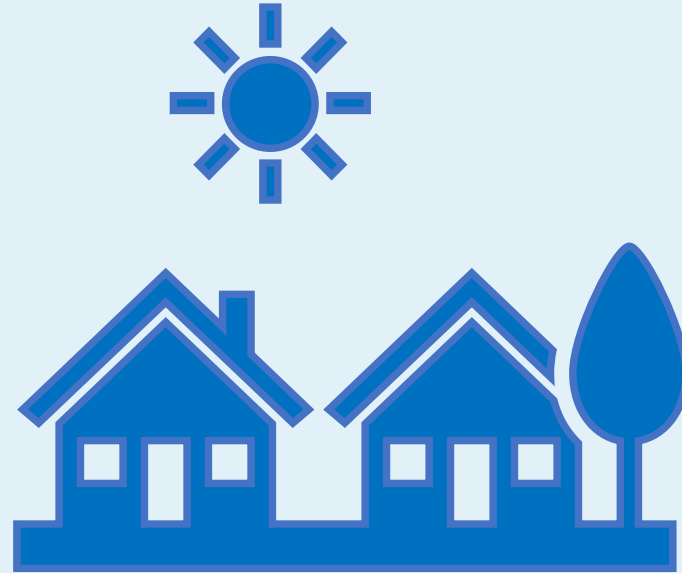


Using
ATNE(approximated
trajectory Nash
equilibrium) and
greedy local decision
rule

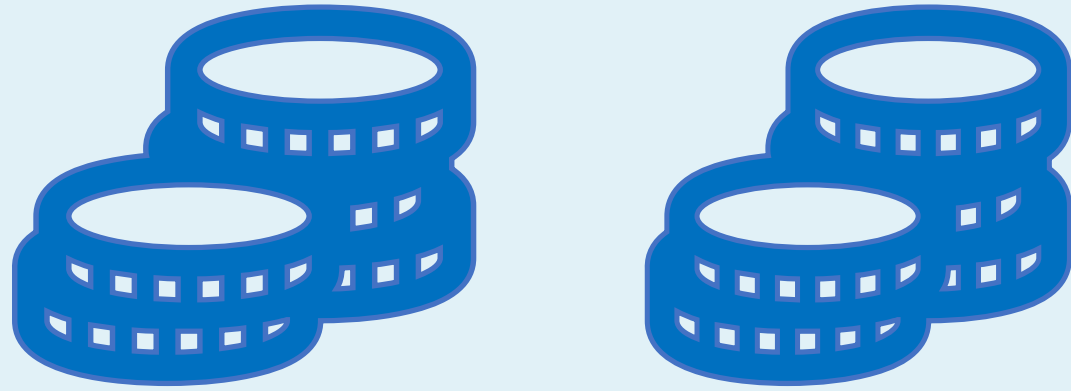
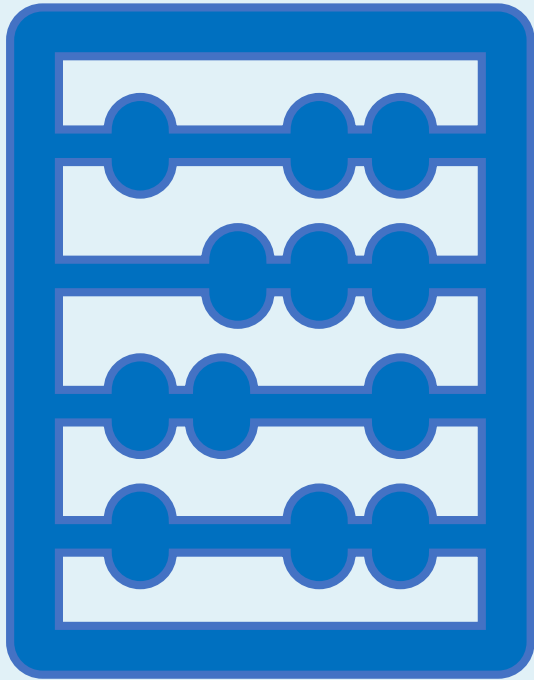


With the greedy local decision rule,
the agents are only looking at their
nearby which helps to cut down on
the time to compute.

With the approximated Nash
equilibrium, the agent is calculating
the movements that are made based
on the locations of those neighbors
only.



Expected utility and Poisson-Binomial approximation



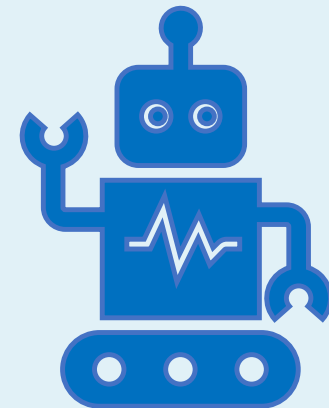
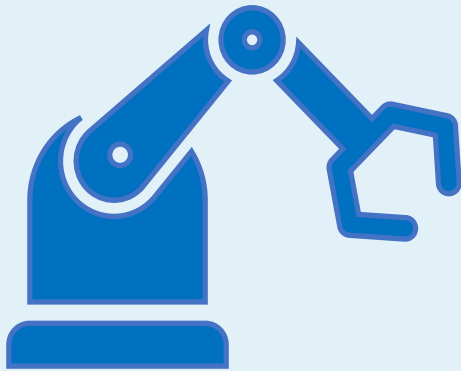
Computing expected utility leads to exponential complexity due to the fact that there are many factors and possible things to calculate.

The Poisson-Binomial approximation helps by reducing the complexity as every participant will have the same cost.

Conceptual AI Links

Multi-agent rational agent system in MAI Comparison

- It is rational as each individual autonomous vehicle is a greedy agent that achieves maximum utility.
- They are all autonomous, and goal oriented for themselves instead of a group.
- Each agent makes an individual decision to maximize their utility in a competitive environment.



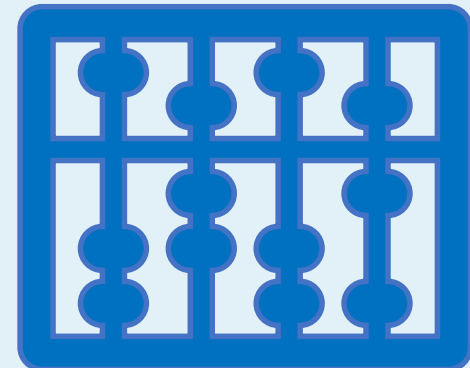
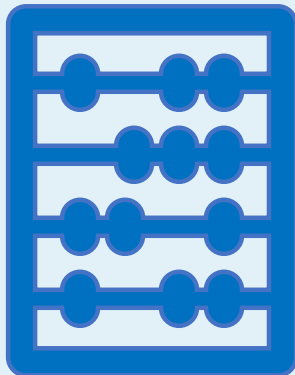
Reinforcement Learning vs The Greedy Expected-utility Approach Agent's Classification

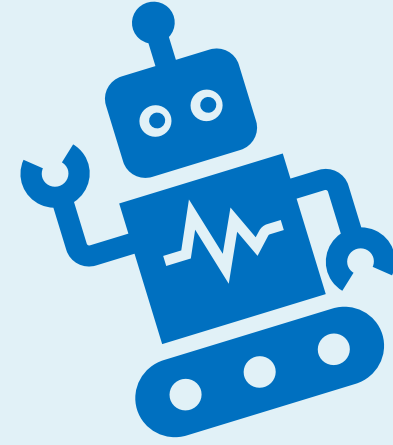
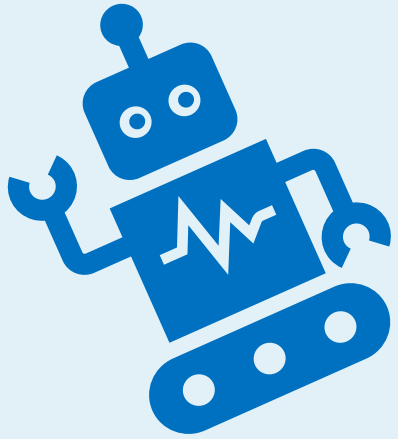
The agent's classification for reinforcement would change the agent to a learning agent.

It will be able to change the approach based on feedback it learned from the system.

If new rules were applied to the game, it would be able to adapt.

The greedy agent cannot change its approach.





Thank you!

